EXPERIMENT:14-Experiment to understand feature scaling.

Description:

The program demonstrates feature scaling using StandardScaler and MinMaxScaler on a sample dataset.

PROGRAM:

```
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler, MinMaxScaler
# New dataset
data = {
    "Study_Hours": [1, 3, 5, 7, 9],
    "Test_Score": [40, 50, 70, 85, 100],
    "Attendance": [55, 65, 80, 90, 100]
}
df = pd.DataFrame(data)
print("Original Dataset:")
print(df)
# Standard Scaling
sc = StandardScaler()
standard_scaled = sc.fit_transform(df)
print("\nStandard Scaled Data:")
print(standard_scaled)
# Min-Max Scaling
mms = MinMaxScaler()
minmax_scaled = mms.fit_transform(df)
print("\nMin-Max Scaled Data:")
```

print(minmax_scaled)

OUTPUT:

```
Original Dataset:
   Study_Hours  Test_Score  Attendance
0            1          40          55
1            3          50          65
2            5          70          80
3            7          85          90
4            9         100         100

Standard Scaled Data:
[[-1.41421356 -1.31818182 -1.41022048]
 [-0.70710678 -0.86363636 -0.79708114]
 [ 0.          0.04545455  0.12262787]
 [ 0.70710678  0.72727273  0.73576721]
 [ 1.41421356  1.40909091  1.34890655]]

Min-Max Scaled Data:
[[0.         0.         0.        ]
 [0.25       0.16666667 0.22222222]
 [0.5        0.5        0.55555556]
 [0.75       0.75       0.77777778]
 [1.         1.         1.        ]]
```